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December 2020

Operation and Maintenance Manual

Caterpillar Machine Fluids Recommendations

All Machines87A 1-UP
(Machine)

PROOF

Special Publication

Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards, including human factors that can affect safety. This person should also have the necessary training, skills and tools to perform these functions properly.

Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.

Do not operate or perform any lubrication, maintenance or repair on this product, until you verify that you are authorized to perform this work, and have read and understood the operation, lubrication, maintenance and repair information.

Safety precautions and warnings are provided in this manual and on the product. If these hazard warnings are not heeded, bodily injury or death could occur to you or to other persons.

The hazards are identified by the “Safety Alert Symbol” and followed by a “Signal Word” such as “DANGER”, “WARNING” or “CAUTION”. The Safety Alert “WARNING” label is shown below.



The meaning of this safety alert symbol is as follows:

Attention! Become Alert! Your Safety is Involved.

The message that appears under the warning explains the hazard and can be either written or pictorially presented.

A non-exhaustive list of operations that may cause product damage are identified by “NOTICE” labels on the product and in this publication.

Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. You must not use this product in any manner different from that considered by this manual without first satisfying yourself that you have considered all safety rules and precautions applicable to the operation of the product in the location of use, including site-specific rules and precautions applicable to the worksite. If a tool, procedure, work method or operating technique that is not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that you are authorized to perform this work, and that the product will not be damaged or become unsafe by the operation, lubrication, maintenance or repair procedures that you intend to use.

The information, specifications, and illustrations in this publication are on the basis of information that was available at the time that the publication was written. The specifications, torques, pressures, measurements, adjustments, illustrations, and other items can change at any time. These changes can affect the service that is given to the product. Obtain the complete and most current information before you start any job. Cat dealers have the most current information available.

NOTICE

When replacement parts are required for this product Caterpillar recommends using original Caterpillar® replacement parts.

Other parts may not meet certain original equipment specifications.

When replacement parts are installed, the machine owner/user should ensure that the machine remains in compliance with all applicable requirements.

In the United States, the maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

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Foreword

Fluids/Filters Recommendation

Literature Information

This manual should be stored in the literature holder or in the literature storage area on the machine. Immediately replace this manual if lost, damaged, or unreadable.

The information contained in this document is the most current information available for fluid maintenance and service products. Special maintenance and service products may be required for some machine compartments. Refer to the Operation and Maintenance Manual for your machine for the maintenance and service requirements. Read, study, and keep this manual with the product. This manual should be read carefully before using this product for the first time and before performing maintenance.

Whenever a question arises regarding your product, or this publication, consult your dealer for the latest available information.

Safety

Refer to the Operation and Maintenance Manual for your machine for all safety information. Read and understand the basic safety precautions listed in the Safety Section. In addition to safety precautions, this section identifies the text and locations of warning signs used on the machine.

Read and understand the applicable precautions listed in the Maintenance and Operation Sections before operating or performing lubrication, maintenance, and repair on this machine.

Maintenance

Refer to the Operation and Maintenance Manual for your machine to determine all maintenance requirements.

Proper maintenance and repair are essential to keep the equipment and systems operating correctly. As the owner, you are responsible for the performance of the required maintenance listed in the Owner Manual, Operation and Maintenance Manual, and Service Manual.

Maintenance Interval Schedule

Use the Maintenance Interval Schedule in the Operation and Maintenance Manual for your machine to determine servicing intervals. Use the service hour meter to determine servicing intervals. Calendar intervals shown (daily, weekly, monthly, etc.) can be used instead of service hour meter intervals if calendar intervals provide more convenient servicing schedules and approximate the indicated service hour meter reading. Recommended service should always be performed at the interval that occurs first.

Under extremely severe, dusty, or wet operating conditions, more frequent lubrication and/or filter changes than is specified in the maintenance intervals chart might be necessary.

Following the recommended maintenance intervals reduces the risk of excessive wear and potential failures of components.

Aftermarket Products and Warranty

NOTICE

When auxiliary devices, accessories or consumables (filters, oil, additives, catalysts, fuel, etc.) made by other manufacturers are used on Cat products, the Caterpillar warranty is not affected simply because of such use. Failures that result from the installation or usage of other manufacturers auxiliary devices, accessories or consumables, however, are not Caterpillar factory defects and therefore are NOT covered by Caterpillar's warranty.

Caterpillar is not in a position to evaluate the many auxiliary devices, accessories or consumables promoted by other manufacturers and their effect on Cat products. Installation or use of such items is at the discretion of the customer who assumes ALL risks for the effects that result from this usage.

Furthermore, Caterpillar does not authorize the use of its trade name, trademark, or logo in a manner which implies our endorsement of these aftermarket products.

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Maintenance Section

Lubricant Specifications

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Lubricant Information

SMCS Code: 1000; 1300; 7581

NOTICE
Every attempt is made to provide accurate, up-to-date information. By the use of this document, you agree that Caterpillar Inc. is not responsible for errors or omissions.

The information that is provided is the latest recommendations for Cat diesel engines that are covered by this Special Publication. This information supersedes all previous recommendations which have been published for Cat diesel engines that are covered by this Special Publication. Special fluids are required for some engines. These fluids will still be necessary in those engines. Refer to the applicable Operation and Maintenance Manual.

This publication is a supplement to the Operation and Maintenance Manual. This publication does not replace the engine-specific Operation and Maintenance Manuals.

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

Failure to follow the recommendations found in this Special Publication can cause engine failures, shortened engine service life, and reduced engine performance.

In order to avoid potential damage to your Cat engine, only purchase Cat fluids and Cat filters through your Cat dealer or Cat authorized outlets. For a list of authorized Cat parts outlets in your area, consult your Cat dealer.

If you purchase what appear to be Cat fluids and/or Cat filters through other outlets/sources, you are at a very high risk of purchasing counterfeit (“look-alike”) products.

Counterfeit or “look-alike” products may visually appear the same as the original Cat product. The product performance and internal quality will typically be very low.

Counterfeit or “look-alike” products have a very high likelihood of causing and/or allowing engine and/or machine compartment damage.

Many of the guidelines, recommendations, and requirements that are provided in this Special Publication are interrelated. Before using the provided information, the user is responsible to read this Special Publication and understand the information provided.

The user is responsible to follow all safety guidelines found in this Special Publication and in the engine Operation and Maintenance Manual when performing all recommended and/or required engine, engine systems, and/or machine maintenance.

For questions concerning the information presented in this Special Publication and/or in your product Operation and Maintenance Manual, and/or for additional guidelines and recommendations (including maintenance interval recommendations/requirements) consult your Cat dealer.

Commercial products that make generic claims of meeting “Cat” and/or “Cat requirements" without listing the specific Cat recommendations and/or requirements that are met may not provide acceptable performance. Reduced engine and/or machine fluid compartment life may result. Refer to this Special Publication and refer to the product Operation and Maintenance Manual for Cat fluids recommendations and/or requirements.

Use of fluids that do not meet at least the minimum performance recommendations and/or requirements may lead to lower compartment performance and/or compartment failure.

Problems/failures that are caused by using fluids that do not meet the minimum recommended and/or required performance level for the compartment are not warrantable by Caterpillar. The fluid manufacturer and customer are responsible.

When fluids made by other manufacturers are used on Cat products, the Cat warranty is not affected simply because of such use. Failures that result from the installation or usage of other manufacturer fluids, however, are not Cat factory defects and therefore are NOT covered by the Cat warranty. Cat is not in a position to evaluate the many fluids promoted by other manufacturers and their effects on Cat products. Installation or use of such items is at the discretion of the customer who assumes ALL risks for the effects that result from this usage.

Different brand oils may use different additive packages to meet the various engine performance category/specification requirements. For the best results, do not mix oil brands.

The overall performance of engine and machine compartments is dependent on the choice of the lubricants and on the maintenance and cleanliness practices. The choices include filtration products, contamination control, tank management, and general handling practices. Cat designed and produced filtration products offer optimal performance and system protection.

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In order to obtain additional information on Cat designed and produced filtration products, refer to Special Publication, SEBU9208, “Caterpillar Filters Recommendations”. Also refer to the “Reference Material” article, “Filters” and “Miscellaneous” topics in this Special Publication. Consult your Cat dealer for assistance with filtration recommendations for your Cat machine.

Note: Inorder to help ensure the maximum expected compartment performance and life, use a fluid that meets Cat highest level of fluid performance as described in this Special Publication for the compartment. Using a fluid that is considered an acceptable, but lower performing option for typical applications, will provide lower performance.

NOTICE
Faulty engine coolant temperature regulators, or operating with light loads, short operation cycles, excessive idling, or operating in applications where normal operating temperature is seldom reached can contribute to excessive water in the crankcase oil. Corrosive damage, piston deposits, increased oil consumption, and other damage can result. If a complete oil analysis program is not followed or if the results are ignored, the potential for damage increases. Follow engine warmup recommendations provided in this Special Publication and/or given in your engine Operation and Maintenance Manual.

Cat Fluids

Cat fluids have been developed and tested by Cat to optimize the performance and the life of Cat engines and machines. The quality of finished oil is dependent on the quality of the base stock, the quality of the additives and the compatibility of the base stock and additives. Cat fluids are formulated of high-quality refined oil base stocks and additives of optimal chemistry and quantity in order to provide high performance in engines and machine components. Cat fluids are used for factory fill of Cat engines and components and are offered by Cat dealers for service fills and as aftermarket products. Consult with your Cat dealer for more information on these Cat fluids.

Cat recommends the use of the following Cat fluids:

Table 1

Cat Lubricants		Viscosity Grade
Diesel Engine Oil-Ultra Low Sulfur (1)(2) (API CK-4)	Cat DEO-ULS	SAE 15W-40
		SAE 10W-30
	Cat DEO-ULS SYN	SAE 5W-40
	Cat DEO Cold Weather	SAE 0W-40

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(Table 1, contd)

Cat Lubricants		Viscosity Grade
Diesel Engine Oil [1] (API CI-4/API CI-4PLUS)	Cat DEO	SAE 15W-40
		SAE 10W-30
Multi-Application Engine Oil	Cat MAEO Plus (Non-Cat mixed fleets)	SAE 10W-40
Pre Combustion Oil	Cat PCO	SAE 40
Transmission/Drive Train Oil	Cat TDTO	SAE 10W
		SAE 30
		SAE 50
	Cat TDTO-TMS	Multigrade
	Cat TDTO Cold Weather	SAE 0W-20
Special Application Transmission Oil	Cat SATO	Multigrade
Automatic Transmission Fluid	Cat ATF-HD2	Multigrade Synthetic
Final Drive and Axles Oil	Cat FDAO	SAE 60
	Cat FDAO SYN	Multigrade
Gear Oil	Cat GO	SAE 80W-90
		SAE 85W-140
	Cat Synthetic GO	SAE 75W-140
	Cat Marine POD GO	SAE 75W-90
	Cat Compactor Oil	ISO 220
	Cat GO	ISO 220
Hydraulic Oil	Cat HYDO Advanced	SAE 10W
		SAE 20W
		SAE 30
	Cat Bio HYDO Advanced (HEES)	ISO 46 Multigrade
Multipurpose Tractor Oil	Cat MTO	SAE 10W-30

(1) Cat engine oils exceed the performance requirements of the respective API categories
(2) These oils have changed from API CJ-4 to API CK-4 early in 2017

Note: Additional Cat fluids may be available. Some specialized oils are available in limited regions globally. Refer to your Cat dealer for oil availability.

Note: Cat offers Special Application Engine Oil (SAEO), of SAE 30 and SAE 40 viscosity grades that fullfills all the requirements of API CF category oil. These oils are recommended for use in 3116 and 3126 marine engines only.

Note: Cat offers a Diesel Engine oil (DEO), SAE 40, that fulfills all the requirements of API CF category for use in Cat 3600 engine models only. Refer to Special Publication, SEBU7003.

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Table 2

Cat Grease products	
Grease	Cat Utility Grease
	Cat Prime Application Grease
	Cat Extreme Application Grease
	Cat Extreme Application Grease - Desert
	Cat Extreme Application Grease - Arctic (Semi-Synthetic)
	Cat Ball Bearing Grease
	Cat White Assembly Grease
	Cat Hammer Paste

Table 3

Cat Engine Coolant Products	
Cat Extend Life Coolants and Extenders	Cat Extended Life Coolant (Cat ELC)
	Cat Extended Life Inhibitor (Cat ELI)
	Cat Extended Life Extender (Cat ELC Extender)
Cat Conventional Coolants and Supplemental Coolant Additives	Cat Diesel Engine Antifreeze Coolant
	Cat Supplemental Coolant Additive (SCA)

Note: Cat fluids availability will vary by region.

This information applies to Cat machines. For more lubricant recommendations see the following Special Publications:

- Special Publication, SEBU6251, “Caterpillar Commercial Diesel Engine Fluid Recommendations”
- Special Publication, SEBU6385, “Caterpillar On-Highway Diesel Engine Fluids Recommendations”
- Special Publication, SEBU6400, “Caterpillar Gas Engine Fluids Recommendations”
- Special Publication, SEBU7003, “Caterpillar 3600 Series and C280 Series Diesel Engine Fluids Recommendations”

Always consult your Cat dealer to ensure that you have the current revision level of the publication.

Note: The optimal application of the lubricants is dependent on the oil quality and the maintenance practices such as contamination control, tank management, and general handling practices.

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Engine Oil
(Cat Machine Diesel Engines)

SMCS Code: 1348; 7581

Engine lubricants play multiple roles in engines. Appropriate lubricants offer the following:

- Provide lubrication to the moving components of the engine under a wide range of temperatures and pressures
- Keep the engine components clean and remove wear debris
- Remove heat from the lubricated components
- Neutralize acidic products due to combustion process
- Protect the engine from cavitation and foaming
- Protect the engine from corrosion and rust
- Control Oil Consumption
- Disperse/Solubilize Contaminants (soot)

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- Support the regulated engine emissions limits

Current lubricant formulations are more advanced and complex than older formulations. Current lubricants are developed to support advanced engine technologies as well as those with lower emissions while supporting the performance and durability of these engines. At the same time, these oils are backwards compatible and protect older engines.

Cat high-performance oils are produced and validated using industry standard tests, internal proprietary tests, field tests, and often prior experience with similar formulations. The American Petroleum Institute (API) categories describe the key industry standards that set the minimum acceptable performance for engine oils. Other global standard setting organizations may also develop common standards, for example the European ACEA oil specs. Cat high quality and high-performance lubricants are validated based on these factors.

Use the recommended engine oils as provided in this special publication to provide optimal engine performance and life and comply with regulated emission reductions. Due to the significant variations in the quality and performance of commercially available oils on a global basis, Cat recommends the use of Cat oils as detailed in this article.

Cat Diesel Engine Oils

Cat fluids have been developed and tested by Caterpillar to provide the full performance and service life that has been designed and built into Cat engines. The quality of finished oil is dependent on the quality of the base stock, the quality of the additives, and the compatibility of the base stock and additives. Cat fluids are formulated of high-quality refined oil base stocks and additives of optimal chemistry and quantity in order to provide high performance in engines and machine components. Cat fluids are used for factory fill of Cat engines and components and are offered by Cat dealers for service fills and as aftermarket products. Consult your Cat dealer for more information on these Cat fluids.

Cat recommends the use of the lubricants given in Tables 1 and 9 in diesel engines covered by this Special Publication.

This information applies to Cat machine engines. For more lubricant recommendations refer to other SEBU publications listed earlier in this Chapter.

Note: The optimal application of the lubricants is dependent on the oil quality and the maintenance practices. Such practices include contamination control, oil tank management and general handling practices.

Cat Diesel Engine Oils Recommendations

Cat DEO-ULS and Cat DEO multigrade oils are the preferred and recommended oils for use in ALL Cat diesel engines that are covered by this Special Publication. Commercial alternative diesel engine oils are, as a group, second choice oils. Refer to Table 5 below for information.

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Table 4

Cat Engine Lubricants Recommendations	
Engines with aftertreatment devices[1]	Required: Cat DEO-ULS (API CK-4)
	Commercial (second-class) oils: API CK-4; Cat ECF-3/API CJ-4
Engines without aftertreatment devices	Recommended oils: Cat DEO-ULS (API CK-4); Cat DEO (API CI-4/ API CI-4 PLUS)
	Commercial (second class) oils: API CK-4; Cat ECF-3/API CJ-4, Cat CF-2/API CI-4

(1) For details, refer to Recommendations for US EPA Tier 4 Certified Nonroad Engines and Recommendations for Europe Stage V Certified Nonroad Engines

Note: Cat DEO-ULS (API CK-4) oil is backwards compatible and can be used in all engines with some restrictions related to fuel sulfur level, refer to Table 4 for more details. Cat DEO (API CI-4/API CI-4 PLUS) can be used in engines that are Tier 3 emissions certified and prior, and in engines that do not use aftertreatment devices.

Caterpillar has released a new Cat diesel engine oil, Cat DEO-ULS that is per the latest API CK-4 heavy duty engine oil category. The new Cat DEO-ULS replaces the prior Cat oil that was per API CJ-4 category, but it is of the same brand name. The new Cat has the following characteristics:

- Improved oxidation stability (per ASTM D8048).
- Improved air release (Per ASTM D8047).
- Improved used oil shear stability per tighter specification limits.
- Preserved the same level of phosphorous, 1000 ppm (parts per million) (mg/kg), to ensure achieving long hour engine durability goals
- Same chemical limits as API CJ-4 oil category, designed for use in engines with aftertreatment devices.

Note: The new API FA-4 Heavy Duty Engine Oil Category is NOT allowed in Cat engines. API FA-4 is a special low High Temperature High Shear (HTHS) viscosity oil that is designed for certain 2017 and later On-Highway engine models.

Note: Each of the Cat ECF specifications provides increased performance over lower Cat ECF specifications. For example, Cat ECF-3 provides higher performance than Cat ECF-2 and Cat ECF-3 provides much higher performance than Cat ECF-1-a. Refer to Table 6 for details.

The engine oils recommended/required for Tier 4 certified engines are formulated with limited ash and chemical limits:

- 1 percent sulfated ash maximum
- 0.12 percent phosphorous maximum
- 0.4 percent sulfur maximum

These chemical limits were developed to support the expected aftertreatment devices life, performance, and service intervals. Use of oils other than those listed in this section of this special publication in aftertreatment equipped engines can negatively impact performance of the aftertreatment devices, can contribute to Diesel Particulate Filter (DPF) plugging and/or can cause the need for more frequent DPF ash service intervals.

Cat DEO-ULS and Cat DEO are recommended for all non-Tier 4 engines that use Ultra Low Sulfur Diesel (ULSD) or Low Sulfur Diesel (LSD) fuels. Cat DEO is recommended for engines using fuels of sulfur levels that exceed 0.2 percent (2000 ppm). Cat DEO-ULS may be used in these applications if an oil analysis program is followed. The oil change interval may be affected by the fuel sulfur level. Refer to Table 6 in this section of this Special Publication.

Cat DEO-ULS is formulated with 1000 ppm (parts per million) (mg/kg) phosphorous level to ensure achieving long hour engine durability goals.

Cat diesel engine oils, DEO-ULS and Cat DEO are rigorously tested with full-scale proprietary Cat engine tests to ensure optimal protection of Cat diesel engines. The tests include the following: sticking of the piston rings, piston deposits, oil control tests, wear tests and soot tests. Proprietary tests help ensure that Cat oils provide superior performance in Cat diesel engines.

There are significant variations in the quality and performance of commercially available oils. For this reason, Cat recommends the Cat oils listed in Table 4 and the guidelines in Table 4.

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NOTICE
Do not use single grade API CF oils or multigrade API CF oils in Cat Series 3500, Series C175and smaller Direct Injection (DI) diesel engines. Single grade API CF oils (or oils that meet all the performance requirements of API CF category) may only be used in Cat Series 3600 and Series C280 diesel engines, and older Cat engines that have precombustion chamber (PC) fuel systems. Oils that are used in Cat Series 3600 and Series C280 diesel engines must also pass a 7000 our field performance evaluation. Consult your Cat dealer for details

Cat diesel engine oils are formulated with the optimal amounts and chemistry of various additives including detergents, dispersants, antioxidants, alkalinity, antifoam, viscosity modifiers, and others. These oils have been proven in internal tests and in the field to provide superior performance in Cat diesel engines where recommended for use.

Cat DEO-ULS and Cat DEO are multigrade oils. Refer to Table 1 in this article for details. Multigrade oils provide the correct viscosity for a broad range of operating temperatures. Multigrade oils provide the appropriate oil film thickness for moving engine components such as piston, ring and liners, bearings, valve train and others.

Mixed fleets: Cat diesel engine oils exceed many of the performance requirements of the corresponding API categories and of other manufacturers of diesel engines. Therefore these oils are excellent choices for many mixed fleets. Refer to the engine manufacturer literature for the recommended categories/specifications. Compare the categories/specifications to the specifications of Cat diesel engine oils. The current industry standards for Cat diesel engine oils are listed on the product labels.

Also, refer to the datasheets for the product for technical details.

Non-EPA certified engines: Cat DEO-ULS and Cat DEO are recommended for all pre-Tier 4 engines that use Ultra Low Sulfur Diesel (ULSD) or Low Sulfur Diesel (LSD) fuels. Cat DEO is recommended for engines using fuels of sulfur levels that exceed 0.2 percent (2000 ppm). Cat DEO-ULS may be used in these applications if an oil analysis program is followed. The oil change interval may be affected by the fuel sulfur level. Refer to the Oil Analysis and Cat S.O.S. Services section of this Special Publication.

Tier-4 certified engines: Use appropriate lubricating oils that are compatible with the engine certification and aftertreatment system and with the fuel sulfur levels. Refer to the oil recommendations for Tier 4 certified engines in this Chapter, to “Diesel Fuel Sulfur Impacts” article of the “Fuels Specifications” section and to “Lubricants Specifications” section of this Special Publication.

Cat DEO-ULS (API CK-4) is compatible for use in engines with aftertreatment devices. Refer to the appropriate section in this article for details.

Consult your Cat dealer for part numbers and for available sizes of containers.

Note: API oil category CF is obsolete. The API (American Petroleum Institute) does not license this category effective end of 2010. API does not validate the quality of API CF oils and does not allow the display of API symbol (also called API doughnut) with CF as highest claim on the oil container. Oils that claim the requirements of API CF can be used ONLY in Cat Series 3112 and Cat Series 3126 Marine Engines and in 3600 engine models per the OMM of these engines. Refer to the details given in the Marine engine section in this Special Publication.

Recommendation for US EPA Tier 4 Certified Nonroad Engines

All diesel engines with aftertreatment devices are REQUIRED to use specially formulated engine oils and specific diesel fuels. The engine categories that are certified per the emissions regulations listed below commonly have aftertreatment devices:

- United States Environmental Protection Agency (EPA) Tier 4 Nonroad
- European Union Stage IIIB, IV, and V, Nonroad
- Japan 2014 (Tier 4) Nonroad
- Korea Tier 4 Nonroad
- India Bharat Stage IV for Construction Equipment Vehicles
- China Nonroad Stage IV

Engine Oils

The REQUIRED engine oils are listed below. These oils are developed with restricted ash level and chemical limits that are suitable for use in engines with aftertreatment.

- Cat DEO-ULS (preferred)
- API CK-4 engine oil category
- Oils meeting the Cat ECF-3 specification
- API CJ-4 engine oil category
- ACEA E9

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Note: Note that ACEA E9 oils are validated using some but not all ECF-3 and API CJ-4 standard engine performance tests. Consult your oil supplier when considering use of an oil that is not API CK-4, Cat ECF-3 or API CJ-4 qualified.

The chemical limits are detailed in the Section titled “Cat Diesel Engine Oils Recommendations, and Cat” of this chapter

Diesel Fuels

The diesel fuels REQUIRED by regulations for use in engines that are certified to nonroad emissions standards listed above and in engines equipped with exhaust aftertreatment systems are.

- United States Ultra Low Sulfur Diesel (ULSD) fuel ≤15 ppm (mg/kg) (0.0015 percent) sulfur
- European ULSD ≤10ppm (mg/kg) (0.0010 percent) sulfur fuel. This fuel is also called “sulfur free”
- Other fuels available around the world that are ≤15 ppm (mg/kg) (0.0015 percent) sulfur

Certain governments/localities and/or applications MAY require the use of ULSD fuel. Consult federal, state, and local authorities for guidance on fuel requirements for your area.

ULSD fuel or sulfur-free diesel fuels are suitable for use in all engines regardless of the engine Tier or Stage.

The fuels listed above have to meet the performance levels detailed in the Fuel Information for Diesel Engines chapter in this Publication. The Fuel Information for Diesel Engines Chapter also includes the pertinent recommendations for biodiesel fuels in the engines certified to non-road emissions standards listed above.

Diesel Exhaust Fluid (DEF)

This fluid MUST be used in engines that are equipped with Selective Catalytic Reduction (SCR) systems. DEF must meet all the requirements detailed in the Exhaust Aftertreatment Fluids Specification chapter in this Special Publication.

Aftertreatment systems may include the following:

- Diesel Particulate Filters (DPF)
- Diesel Oxidation Catalysts (DOC)
- Selective Catalytic Reduction (SCR)
- Lean NOx Traps (LNT)

Other systems may apply.

Regulations may vary around the world. Follow all the local regulations and fluids requirements in your area. Refer to your engine-specific Operation and Maintenance Manual, and refer to your aftertreatment device documentation, if available, for additional guidance.

NOTICE
Do not add new engine oil, waste engine oil or any oil product to the fuel unless the engine is designed and certified to burn diesel engine oil (for example Caterpillar ORS designed for large engines). Caterpillar experience has shown that adding oil products to Tier 4 engine fuels (U. S. EPA Tier 4 certified), to Euro IV/ Stage IV certified engine fuels, or to the fuels of engines equipped with exhaust aftertreatment devices, will generally cause the need for more frequent ash service intervals and/or cause loss of performance. Adding oil products to the fuel may raise the sulfur level of the fuel and may cause fouling of the fuel system and loss of performance.

Recommendation for Europe Stage V Certified Nonroad Engines

All the recommendations and requirements given in “Recommendation for US EPA Tier 4 Certified Nonroad Engines” section are applicable to the Europe Stage V type-approved Nonroad Engines

Additionally, for the correct operation of the engine in order to maintain the gaseous and particulate pollutant emissions of the engine within the limits of the type-approval, unless specified otherwise in the engine-specific Operation and Maintenance Manual, **EU Stage V** regulations REQUIRE the diesel fuels (also called non-road gas oil) used in engines operated within the European Union (EU) to have the characteristics below

- The sulfur content should not be greater than 10 mg/kg (20 mg/kg) at point of final distribution
- The Cetane number should not be less than 45
- The biodiesel (also called Fatty Acid Methyl Ester (FAME)) content should not be greater than 8 % volume/volume

Note: Certain Cat engines that are certified per EU Stage V can use up to B20 biodiesel blends. Refer to your engine specific Operation and Maintenance Manual.

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Follow all the local regulations and fluids requirements in your area. Refer to your engine-specific Operation and Maintenance Manual, and refer to your aftertreatment device documentation, if available, for additional guidance.

Commercial Engine Oil

Note: Non-Cat commercial oils are, as a group, second choice oils. Within this grouping of second choice oils there are tiered levels of performance.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

There are significant variations in the quality and performance of commercially available oils. For this reason, Cat recommends the Cat oils listed in Tables 1 and 3.

Caterpillar strongly recommends the use of Cat engine oils in Cat engines since these oils are developed and optimized for use in these engines.

When the recommended Cat diesel engine oils are not used, commercial oils that are API CK-4 licensed and/or meet the requirements of the Cat ECF-1-a, Cat ECF-2, and/or the Cat ECF-3 specification are acceptable, but second choice, for use in Cat diesel engine.

Oils that are API CJ-4, API CI-4/CI-4 PLUS and API CH-4 licensed, and do not meet any Cat ECF specification are, as a group, third choice.

Cat diesel engine oils exceed many of the performance requirements of Cat ECF specifications and of API categories.

Oils that meet only API CI-4/CI-4 PLUS and/or API CH-4 categories and that have not met the requirements of at least one Cat ECF specification may cause reduced engine life.

Note: API FA-4 oils are NOT allowed for use in Cat engines. These oils are designed for use in certain 2017 On-Highway engine models.

API CK-4 oils exceed the performance requirements of prior API categories. Cat developed the Engine Crankcase Fluid (ECF) specifications to ensure the availability of minimum acceptable performance commercial diesel engine oils. The three Cat ECF specifications: Cat ECF-1-a, Cat ECF-2 and Cat ECF-3 are described in Table 5. These specifications require more engine tests than the corresponding API categories.

Each higher Cat ECF specification provides increased performance over lower Cat ECF specifications. For example, Cat ECF-3 provides higher performance than Cat ECF-2 and Cat ECF-3 provides much higher performance than Cat ECF-1-a. Refer to Table 5 for details.

Table 5 below provides details of Cat ECF specifications.

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Table 5

Cat Engine Crankcase Fluids (ECF) Definitions	
Cat Minimum Performance Requirements for Commercial Oils	**Cat ECF Specifications Requirements**
(1)	API CK-4 Oil Category performance requirements
Cat ECF-3	API CJ-4 Oil Category performance requirements
Cat ECF-2	API CI-4 / CI-4 PLUS Oil Category performance requirements and Passing standard Cat C13 engine test per API requirements and sulfated ash ≤ 1.50 percent
Cat ECF-1–a	API CH-4 Oil Category performance requirements and: For oils that are between 1.30 percent and 1.50 percent sulfated ash, passing one additional Cat 1P SCOTE test (“ASTM D6681”) is required and sulfated ash ≤ 1.50 percent

(1) Caterpillar did not develop an ECF external specification for API CK-4. Cat DEO-ULS API CK-4 oil is specifically developed and validated for Cat engines. The API categories define the minimum common OEM requirements for engine oils

Refer to “Commercial Engine Oil Recommendations” and “The Current American Petroleum Institute (API) Oil Categories” sections of this chapter for information on the API categories and corresponding Cat engine oils.

Note: Cat diesel engine oils are required to pass proprietary full-scale diesel engine testing. The testing is above and beyond the testing required by the various Cat ECF specifications and by the various API oil categories that are also met. This additional proprietary testing helps ensure that Cat multigrade diesel engine oils, when used as recommended, provide superior performance in Cat diesel engines. If Cat diesel engine oils are not used, use only commercial oils that meet the recommendations and requirements stated in this section of this Special Publication.

Note: For engines that are Tier 4 EPA certified, refer to the Recommendations for Tier 4 Engines article in this Engine Oil section. Tier 4 EPA certified engines require specifically formulated oils.

The fuel sulfur level impacts oil drain intervals and hence the choice of oils. Fuels of sulfur level of >0.2 percent (2000 ppm) require the use of oils of adequate ash content such as Cat DEO oils. Cat DEO-ULS is of limited ash content and can be used if an oil analysis program is followed.

In selecting oil for any engine application, both the oil viscosity and oil performance category/specification as specified by the engine manufacturer must be defined and satisfied. Using only one of these parameters will not sufficiently define oil for an engine application.

In order to make the proper diesel engine oil viscosity grade choice, refer to the “Lubricant Viscosities for Ambient Temperatures” table in this Special Publication.

NOTICE
Failure to follow these oil recommendations can cause shortened engine service life due to deposits and/or excessive wear.

NOTICE
Do not use single grade API CF oils or multigrade API CF oils in Cat Series 3500, Series C175and smaller Direct Injection (DI) diesel engines. Single grade API CF oils (or oils that meet all the performance requirements of API CF category) may only be used in Cat Series 3600 and Series C280 diesel engines, and older Cat engines that have precombustion chamber (PC) fuel systems. Oils that are used in Cat Series Series 3600 and Series C280 diesel engines must also pass a 7000 hour field performance evaluation. Consult your Cat dealer for details.

The current API categories represent the minimum requirements for diesel engine oils. These categories are developed in collaboration of OEMs and the Oil and Chemical industries.

The current licensed oils are:

- API CK-4, released December 2016
- API CJ-4, released 2006
- API CI-4, API CI-4PLUS released 2002
- API CH-4, released 1998

Each API category is technically more advanced and of higher performance than the prior one.

Refer to API 1509 document and/or ASTM D4485 for details of the API tests requirements and limits.

PROOF

Note: Obsolete API oil categories are not licensed by the API and hence are of uncontrolled quality. These oils are technically inferior to current oils and can result in reduced engine performance and life. These obsolete oil categories are not allowed in Cat engines.

API FA-4 is **NOT** recommended for Cat engines. API FA oil is designed with low High Temperature High Shear (HTHS) viscosity oil for use in certain 2017 On-Highway truck engines that have to meet certain on-road emissions regulations. This category is not backwards compatible.

In order to help protect your engine, and in order to help optimize oil drain intervals for engine applications and duty cycles, use Cat S·O·S Services oil analysis. Refer to the section below and to the section on Cat S·O·S Services given later in the Special Publication.

NOTICE
Depending on application severity and localized environmental conditions, and also depending on maintenance practices, operating Direct Injection (DI) diesel engines and operating PC (Precombustion Chamber) diesel engines on fuel with sulfur levels over 0.1 percent (1000 ppm) may require significantly shortened oil change intervals in order to help maintain adequate wear protection. Refer to this Special Publication, “Fuel Specifications” section, “Diesel Fuel Sulfur” topic for additional information.

Fuel Sulfur Impact on Engine Oil

The use of Cat S·O·S Services oil analysis is recommended for determining oil life.

For Caterpillar machines operating under emissions regulations such as US EPA Tier4, EU Stage V or other emission regulations controls, the maximum fuel sulfur level permitted by regulations in the US is 0.0015% or 15 ppm. Maximum permitted fuel sulfur levels in other emissions regulated countries may vary from .0015% or 15 ppm slightly and must be followed.

In order to help protect your engine and optimize oil drain intervals for engine applications and duty cycles, it is important to use SOS Services oil analysis to determine if the sulfur has degraded the oil. **Use Cat S·O·S Services oil analysis per the following general guidelines:**

- Recommended normally
- For fuel sulfur level up to 0.05 percent (500 ppm), no additional sampling is required. Follow the recommended oil sampling in the machine OMM
- For fuel sulfur level of > 0.05 percent to 0.5 percent (500 ppm to 5000 ppm), SOS services oil analysis is strongly recommended to determine oil drain intervals. Sample the oil every 250 hours until a trend is established, then sample as needed.
- For fuel sulfur level of > 0.50 percent (>5000 ppm), SOS services oil analysis is required to determine oil drain intervals. Sample oil every 125 hours until a trend is established, then sample as needed.

These recommendations apply for Cat DEO-ULS and DEO oils.

Note: Engine operating conditions play a key role in determining the effect that fuel sulfur will have on engine deposits and on engine wear. Refer to the Diesel Fuel Chapter in this Special Publication for the impacts of high sulfur in the fuel on engine performance and oil service interval and consult your Cat dealer for guidance when fuel sulfur levels are above 0.2% (2000 ppm).

The engine oil Total Base Number (TBN) and ash content can impact the oil performance and the oil service interval. Excessive piston deposits can be produced by oil with a high TBN and/or high ash. These deposits can lead to a loss of control of the oil consumption and to the polishing of the cylinder bore. The engine oil TBN can be measured by SOS Services upon special request. However, analysis of used oil for parameters such as oxidation, wear metals and other typical S·O·S Services analyses are strong indicators of the condition of the oil and impacts of fuel sulfur level on its degradation.

Refer to Severe Operation Conditions in this Section of this Special Publication.

Consult a trained S·O·S Services analyst when making oil drain decisions based on oil sample results.

Note: Do NOT use only this Special Publication as a basis for determining oil drain intervals.

This Special Publication does not address recommended oil drain intervals, but rather provides guidance that should be used with your specific engine/machine Operation and Maintenance Manuals in determining acceptable oil drain intervals.

Consult your engine/machine Operation and Maintenance Manuals, and consult your Cat dealer for additional guidance, including but not limited to guidance on establishing optimized and/or acceptable oil drain intervals.

Note: The use of Cat S·O·S Services oil analysis helps environmental sustainability as the best way to optimize oil life, and help engines reach expected life.

PROOF

Consult your Cat dealer regarding the testing required to establish a safe, optimized oil drain interval.

Severe Applications

An engine which operates outside of normal conditions is operating in a severe service application.

An engine that operates in a severe service application may need more frequent maintenance intervals in order to maximize the following conditions:

- Reliability
- Service life

Severe Applications Require the Use of Higher Performing Diesel Engine Oils. Examples of severe duty applications include, but not limited to the following:

- operating at greater than 75% load factor
- operating in high humidity
- operating with fuel sulfur levels that are above 0.2% (2000 ppm)

In order to help ensure the maximum expected compartment performance and life, higher performing fluids as described in this Special Publication may be required. Fluids that meet only the minimum performance levels may result in reduction of maintenance intervals. For the maximum expected engine compartment performance and life and where LSD or ULSD fuels are available, Cat DEO-ULS or, as a second choice, oils meeting the API CK-4 or Cat ECF-3 specification must be used.

The vast number of varying applications of Cat machines and engines makes it impossible to identify all the factors which may contribute to severe service operation. Consult your Cat dealer for the unique maintenance that may be necessary for your engine.

An application is a severe service application if any of the following conditions apply:

Severe Environmental Factors

- Frequent operation in dirty air
- Frequent operation at an altitude which is above 1525 m (5000 ft)
- Frequent operation in ambient temperatures which are above 32° C (90° F)
- Frequent operation in ambient temperatures which are below 0° C (32° F)

Severe Operating Conditions

- Frequent operation with inlet air which has a corrosive content
- Operation with inlet air which has a combustible content
- Operation which is outside of the intended application
- Operation with a plugged fuel filter
- Extended operation at low idle (more than 20% of hours)
- Frequent cold starts at temperatures below 0° C (32° F)
- Frequent dry starts (starting after more than 72 hours of shutdown)
- Frequent hot shutdowns (shutting down the engine without the minimum of 2 minutes to 5 minutes of cool down time)
- Operation above the engine rated speed
- Operation below the peak torque speed
- Operating with fuel which does not meet the standards for distillate diesel fuel as stated in Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” “Distillate Diesel Fuel”
- Operating with a blend of distillate fuel which contains more than 20 percent (B20) biodiesel

Improper Maintenance Procedures (Maintenance Procedures Which May Contribute to a Severe Service Application)

- Inadequate maintenance of fuel storage tanks from causes such as excessive water, sediment, and microorganism growth.
- Extending maintenance intervals beyond the recommended intervals
- Using fluids which are not recommended in Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations”
- Extending maintenance intervals for changing the engine oil and engine coolant without S·O·S validation
- Extending maintenance intervals for changing air filters, oil filters, and fuel filters
- Failure to use a water separator

PROOF

- Using filters which are not recommended by Special Publication, PEWJ0074, “2008 Cat Filter and Fluid Application Guide” and/or “Caterpillar Filters Recommendations” (SEBU 9208)
- Storing the engine for more than 3 months but less than 1 year (For information about engine storage, refer to Special Publication, SEHS9031, “Storage Procedure for Caterpillar Products”)

For fuel and coolant analysis and use of SOS services, refer to the Diesel Fuel and Coolants chapters in this Special Publication.

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Hydraulic Oil

SMCS Code: 5095; 7581

Applications

- Hydraulic Systems
- Hydrostatic Transmissions

Caterpillar has developed a new class of hydraulic oils, the HYDO Advanced fluids. These new advanced fluids are designed for optimal performance in new advanced hydraulic systems that are designed with close tolerances for precise operation and that operate under demanding conditions of temperatures and pressures. These fluids ensure optimal performance in current and previous hydraulic systems. Four new products have been introduced:

- HYDO Advanced 10, SAE 10W
- HYDO Advanced 20, SAE 20W
- HYDO Advanced 30, SAE 30
- Bio HYDO Advanced, Multigrade

Cat HYDO Advanced Hydraulic Oils, SAE 10W, SAE 20W, and SAE 30

Cat HYDO Advanced hydraulic oils are designed to provide extended oil drain intervals and to provide extra protection to Cat hydraulic system components and hydrostatic transmissions.

Cat HYDO Advanced hydraulic oils are formulated of premium additives and premium base stocks that pass severe qualification testing requirements, in the field and in the laboratory, to provide superior protection for Cat hydraulic systems.

Cat HYDO Advanced hydraulic oils are preferred in most hydraulic systems and in most hydrostatic systems. Refer to this Special Publication, “Lubricant Viscosities”, Table “Hydraulic Systems Lubricant Viscosities for Ambient Temperatures” for the ambient operating temperatures for Cat HYDO Advanced hydraulic oils.

Cat HYDO Advanced oils allow 6000 hours or higher oil drain intervals for most applications. S·O·S Services oil analysis is recommended when the oil drain interval is increased to 6000 hours or higher. In comparison, non-Cat commercial hydraulic oils (second choice oils), allow for a 2000 hour oil drain interval. Caterpillar recommends following the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your machine. Consult your Cat dealer for details.

Cat HYDO Advanced hydraulic oils offer the following benefits:

- Enhanced anti-wear protection: protect wear surfaces and reduce component wear in various hydraulic pumps.
- Protection against rusting and corrosive wear.
- Dispersion of water prevents pump damage due to water ingestion and prevents system damage due to freezing of water.
- 250% increase in oxidation stability per “ASTM D943”.
- Rapid air release protects against cavitation.
- Improved filtration properties help support component cleanliness.

Cat HYDO Advanced hydraulic oils should be used to achieve maximum life and maximum performance from hydraulic system components and from hydrostatic transmissions. To gain the most benefit from the improved performance designed into Cat HYDO Advanced oils, when switching to Cat HYDO Advanced oils, cross contamination with the previous hydraulic oil should be kept to less than 10%.

If a different viscosity is required due to ambient temperatures, the following Caterpillar oils can be used:

- Cat DEO-ULS and Cat DEO
- Cat DEO-ULS SYN and Cat DEO SYN
- Cat DEO Cold Weather
- Cat TDTO
- Cat TDTO Cold Weather
- Cat TDTO-TMS

PROOF

- Cat MTO

Note: Oil drain intervals of the oils listed above are less than the intervals of Cat HYDO Advanced oils. The oil drain intervals of the listed oils are typically 2000 hours and up to a maximum of 4000 hours. An exception is Cat TDTO Cold Weather oil, which allows for a 6000 hour oil drain interval or higher. S·O·S Services oil analysis is required when the oils listed above are used in Cat hydraulic system components and hydrostatic transmissions.

Note: Caterpillar oil availability will vary by region.

Fire resistant hydraulic Fluids

For applications requiring the use of fire-resistant hydraulic fluids, Caterpillar recommends the use of water-free hydraulic fluids that are Per "DIN 24317 and "ISO 12922" specifications. These fluids include HFDR, HFDU (ester based), HFDU (anhydrous glycol based) and others per the specs given above. Water-based hydraulic fluids are not recommended for use in Cat machine hydraulic systems. The acceptable water-free, fire-resistant hydraulic fluids are available as ISO 46 and 68 viscosity grade products. For more information and questions regarding compatibility and conversion, refer to the product information provided by manufacturers of fire-resistant hydraulic fluids.

Food grade hydraulic fluids

For applications requiring the use of food grade hydraulic fluid such as for machines operating on cattle farms, food processing facilities or others, Caterpillar recommends the use of NSF H1 registered hydraulic fluid that offers high performance and protects the hydraulic system under severe operating conditions. Food Grade fluids may have to comply with the Food and Drug Administration requirements. Refer to the regulations and requirements in your region. For information regarding compatibility, conversion and application details, refer to the product information provided by the manufacturer of your Food-Grade hydraulic fluid.

Caterpillar strongly recommend that the hydraulic oil user avoid any exposure and make every effort to avoid any contact of the hydraulic fluid with the feed supply, even if food grade oils are used.

Cat Bio HYDO Advanced (Hydraulic Oil)

Cat Bio HYDO Advanced is biodegradable non-toxic hydraulic oil that is recommended for use in hydraulic systems when environmental compliance is required or desired. Cat Bio HYDO Advanced has a renewable content that exceeds 90% and complies with the European Eco-Label. Consult Federal, state, or local authorities for guidance on hydraulic oils environmental requirements in your area.

Cat Bio HYDO Advanced is formulated with premium additives and synthetic biodegradable base oil that pass severe qualification testing requirements to provide superior protection for Cat hydraulic systems and hydrostatic transmissions.

Cat Bio HYDO Advanced oils allow 6000 hours or higher oil drain intervals for most applications. S·O·S Services oil analysis is recommended when the oil drain interval is increased to 6000 service hours or higher. In comparison, non-Cat commercial biodegradable hydraulic oils (second choice oils) allow for a maximum 2000 hour oil drain interval. Follow the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your machine. Consult your Cat dealer for details.

Cat Bio HYDO Advanced performs similar to Cat HYDO Advanced 10 and offers the following benefits: superior protection against mechanical wear, foaming, and corrosion in hydraulic systems and in hydrostatic transmissions. This oil should be used to achieve maximum life and maximum performance from hydraulic system components and from hydrostatic transmissions. Cat Bio HYDO Advanced is preferred in most hydraulic systems and in most hydrostatic systems when ambient temperature is between −30° C (−22° F) and 45° C (113° F). Consult your Caterpillar dealer for details.

To gain the most benefit from the improved performance designed into Cat Bio HYDO Advanced oils, when switching to Cat Bio HYDO Advanced oils, cross contamination with the previous hydraulic oil should be kept to less than 3%. Flushing of the hydraulic system is recommended to reduce the cross contamination.

Cat Bio HYDO Advanced can be used in hydraulic systems of other Original Equipment Manufacturers (OEM). Refer to your manufacturer requirements for application details.

PROOF

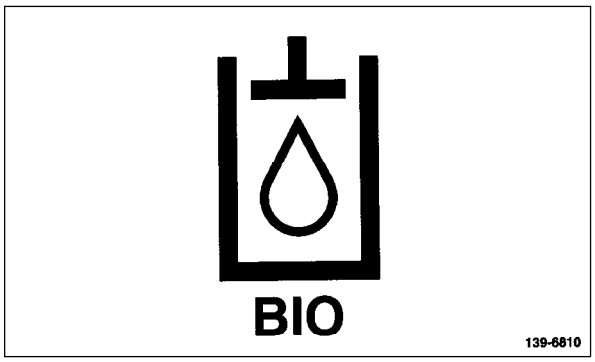


Illustration 1 g00103844

Machines with this symbol are filled with biodegradable hydraulic oil. This symbol is on the hydraulic tank.

Commercial Oils

Note: Non-Caterpillar commercial oils that are acceptable for use in most Cat hydraulic and hydrostatic transmission systems are as a group third choice oils. Within this grouping of third choice oils there are tiered levels of performance.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

The minimum viscosity for commercial alternative oils used in most Caterpillar machine hydraulic systems and in most Cat hydrostatic transmission systems is 6.6 cSt at 100 °C (212 °F) “ASTM D445”.

If Caterpillar oils cannot be used, oils meeting the following commercial specifications can be used in most Cat hydraulic systems and in most Cat hydrostatic transmission systems:

- Engine oils that meet the Cat ECF-1-a, Cat ECF-2, or Cat ECF-3 specifications and have a minimum zinc additive of 0.09 percent (900 ppm)
- Biodegradable oils that meet the Cat BF-2 specification
- TO-4 specification oils that have a minimum zinc additive of 0.09 percent (900 ppm)

Note: Industrial hydraulic oils are not recommended for service fill in Caterpillar hydraulic systems.

Commercial Biodegradable Hydraulic Oil

Note: Non-Cat commercial oils that are acceptable for use in most Cat hydraulic and hydrostatic transmission systems are as a group third choice oils. Within this grouping of third choice oils there are tiered levels of performance.

Commercial oils that meet the Cat BF-2 specification should be used.

Note: Oil meeting the Cat BF-2 specification should only be used in hydraulic systems where specifically recommended.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

If Cat oils cannot be used, commercial biodegradable hydraulic oils that meet the Cat BF-2 Performance Requirements should be used. Cat BF-2 replaces Cat BF-1 Performance Requirements. Cat BF-1 Performance Requirements is obsolete.

NOTICE
Do not use commercial oils that meet the BF-2 specification in compartments containing friction material except Hydraulic Excavators. Refer to the machine Operation and Maintenance Manual or consult your local Cat dealer for guidance before using commercial oils that meet the BF-2 specification in Hydraulic Excavators.

Biodegradable hydraulic oils are formulated with the following components: a saturated HEES (Hydraulic Environmental Ester Synthetic), synthetic base stock, and selected additives. Consult Federal, state, or local authorities for guidance on hydraulic oil environmental requirements in your area.

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Transmission/Drive Train Oil

SMCS Code: 3080; 4000-OC; 4070; 7581

Transmission/drive train oils are classified by the Cat TO-4 and the Cat TO-4M Performance Requirements. These Performance Requirements were developed by Caterpillar for use in Cat Powershift transmissions and in most Cat final drives. Refer to the “Maintenance Section”, “Caterpillar Fluids” article for more information on Caterpillar fluids and to the Tables in the Lubricants Viscosity section for the application of these fluids in various Cat machines.

PROOF

NOTICE
The Cat CD/TO-2 transmission/drive train oil specification has been obsolete since 1990. The obsolete CD/TO-2 specification is tied ONLY to the also obsolete API CD oil category. The “TO-2” oil specification existed only as CD/TO-2. Specification claims of CF/TO-2, ATF/TO-2, Cat TO-2 and other similar “TO-2” claims are not accurate. Caterpillar does not monitor or support the obsolete Cat CD/TO-2 specification. Use of CD/TO-2 oils, or the use of oils claiming to meet “TO-2” in machine compartments where Cat TO-4 specification oils are required is NOT recommended and their use is at the user's risk.

Applications

Transmission/Drive Train oils are recommended for use in Powershift Transmissions, certain Direct Drive Transmissions, certain Final Drives, Differentials and Final Drives, Drive Axles, Transfer Drives, hydrostatic drive/hydraulic and certain Electric Drive systems of various machines.

The details of the applications of Transmission/Drive Train oils are given in the “Lubricant Viscosity” section of this Special Publication.

Cat TDTO (Transmission/Drive Train Oil)

Cat TDTO exceeds the requirements of the Cat TO-4 transmission/drive train oil specification by passing five more Cat tests.

Cat TDTO is balanced to give maximum frictional material life in powershift transmissions. Cat TDTO also helps eliminate brake chatter in wet brake applications in Cat machines. This oil passes the requirements for the TO-4 oil specification which includes the frictional requirements and gear wear requirements. This oil is offered in several lubricant viscosity grades.

Cat TDTO-TMS (Transmission/Drive Train Oil-Transmission Multi-Season)

Cat TDTO-TMS exceeds the requirements of the Cat TO-4M transmission/drive train oil Performance Requirements by passing additional Cat tests.

Cat TDTO-TMS is synthetic multigrade oil that is developed to provide optimal performance in transmissions, wet brakes, final drives, and hydraulic compartments that must operate in wide range of temperatures. Cat TDTO-TMS can be used in Cat machine compartments where Cat TO-4 and TO-4M oils are recommended. This oil can be used in other manufacturer machines that recommend TO-4 or TO-4M Performance Requirements oils.

TDTO-TMS is specifically formulated to reduce transmission gears wear, improve the performance and efficiency of transmissions in cold weather, especially for machines with electronic controls, and to ensure long life and excellent performance for gears, bearings, and friction disc materials.

NOTICE
Cat Transmission oils are formulated for transmissions and drive trains only, and should not be used in engines. Shortened engine life will result.

NOTICE
Do not use the Cat GO (Gear Oil) or commercial gear oil in the machine compartments unless specifically recommended by Caterpillar. The gear oil can cause seals to fail. The seals can also leak. The gear oil may not be compatible with friction materials. The oil can reduce the efficiency of the transmission and the brake performance.

Cat TDTO-Cold Weather

Cat TDTO Cold Weather is a synthetic lubricant blend that provides superior performance and protection in transmissions, torque converters, final drives, hydraulics, and/or wet brakes at temperatures down to -40°F (-40°C). The special formulation creates optimal friction performance and controls transmission slippage while eliminating clutch glazing. Cat TDTO Cold Weather has excellent compatibility with all materials used in Cat machines, including friction materials, seals, and elastomers. The outstanding thermal stability and oxidation resistance gives Cat TDTO Cold Weather long drain intervals.

Commercial Transmission/Drive Train Oils

Note: Non-Caterpillar commercial oils are as a group typically second or third choice oils. Within this grouping of second and/or third choice oils there are typically tiered levels of performance.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

Commercial oils that are substituted for the Cat TDTO must comply with the requirements of the Cat TO-4 or the Cat TO-4M transmission/drive train oil Performance Requirements.

PROOF

Note: Commercial multigrade oils must meet the requirements of the Cat TO-4M specification to be used in transmissions. Multigrade oils that use high molecular weight polymers as viscosity index improvers may lose the viscosity effectiveness. This effectiveness may be lost by the temporary shear of the polymer viscosity index improver. Permanent shear of the polymer viscosity index improver may also occur. These types of multigrade oils are not recommended for Caterpillar drive train compartments. The Cat TO-4M Performance Requirements include a test for the shear stability of multigrade oil.

Cat SATO (Special Application Transmission Oil)

Cat Special Application Transmission Oil (SATO) is multi-grade, semi-synthetic oil. This new Cat oil is introduced for use in all medium wheel loaders for factory fill and service fill.

Cat SATO is applicable in all Medium Wheel Loader, including but not limited to the following models:

- 966K XE, 966M XE, 972M XE (Advanced Power Train Transmission)
- 950M, 962M, 966M, 972M, 980M, 982M (Powershift Transmission)

Note: Cat SATO should not be used in lower power train components including final drives, axles, and differentials.

Cat SATO offers the following benefits:

- Excellent cold start performance.
- Improved oxidation stability compared to the semi-synthetic base stocks.
- Extended oil drain interval for powershift transmissions. The oil drain interval is 2000 hrs for SATO compared to 1000 hrs for TDTO 10 or 30.

For the wheel loaders filled with SATO, topping off with other products from the Cat TDTO family should be limited to less than 10% maximum of sump capacity. The use of higher viscosity fluids for top-off can adversely affect cold start performance. When switching from Cat SATO to Cat TDTO or from Cat TDTO to Cat SATO, no special compartment flushing or alternative procedures are required. Follow standard oil change procedures as outlined in the machine Operation and Maintenance Manual.

Final Drive and Axle Oil

Final Drive and Axle Oils are classified by the Cat FD-1 (Final Drive - 1) oil Performance Requirements. The Cat FD-1 Performance Requirements was developed by Caterpillar for use in certain highly loaded Cat final drives and axles that do **not** contain friction material.

Cat FDAO (Final Drive and Axle Oil), Cat FDAO SYN, or commercial FD-1 are the preferred oil types to maximize gear and bearing life in machine compartments where recommended for use.

Applications

Final Drive and Axle oils are recommended for use in certain Final Drives, Axles, and Differentials. The details of the applications of these oils are given in the “Lubricant Viscosity” section of this Special Publication.

Cat FDAO (Final Drive and Axle Oil)

Cat FDAO is blended specifically for applications with high load and high temperature conditions. In these conditions, the protection of gears and bearings is a primary concern. Cat FDAO is the preferred lubricant for Cat final drives and axles that formerly specified Cat TO-4 oils and that do not contain friction materials. This oil should not be used in compartments that contain friction material unless Cat FDAO or oil meeting Cat FD-1 (Final Drive oil Performance Requirements - 1) has been specified because these oils do not develop a sufficient friction coefficient to satisfy the requirements of most clutches and brakes.

The performance of the Cat FDAO equaled or exceeded the performance of Cat TDTO in 18 of 18 tests for the following characteristics:

- Gear and bearing protection
- Physical properties and chemical properties

Cat FDAO also has the following qualities:

- Compatible with the Cat TDTO oils and the seals for the related compartments
- Excellent protection against rust

PROOF

- Excellent protection against copper corrosion
- Minimizes foaming
- Extended life of the oil due to low rate of oxidation
- Reduced downtime due to extended oil change intervals

Cat FDAO SYN

Cat FDAO SYN is a full synthetic, natural multigrade (non-viscosity improved) final drive and axle oil that is designed for operation in a wide temperature range. Cat FDAO SYN is the preferred oil for extremely cold or hot environments. Cat FDAO SYN exceeds the requirements of Cat FD-1 specification.

Cat FDAO SYN is the preferred lubricant for Cat off-highway truck differentials, front wheels, and final drives that operate under high load factors and/or that are operated continuously (multiple shifts per day).

Cat FDAO SYN is formulated for optimizing component life and for extending the oil change intervals. Cat FDAO SYN use in off-highway truck final drives offers the following:

- Extends the oil drain intervals for off-highway truck final drives as compared to Cat FDAO SAE 60 and FD-1 oils as well as Cat TDTO and TO-4 oils
- Increases bearing life and gear life compared to the use of Cat FDAO SAE 60 and FD-1 oils as well as Cat Cat TDTO or TO-4 oils

The S·O·S oil analysis program is recommended when extending the oil drain intervals.

Commercial Final Drive and Axle Oil

Commercial oils that are substituted for Cat FDAO must comply with the Cat FD-1 final drive oil Performance Requirements.

Note: Non-Caterpillar commercial oils that meet the Cat FD-1 Performance Requirements are second choice oils.

Cat FDAO SYN is formulated for optimizing component life and for extending the oil change intervals. Cat FDAO SYN use in off-highway trucks final drives offers the following:

- Extends the oil drain intervals for off-Highway Trucks final drives as compared to Cat TDTO and TO-4 oils
- Increases bearing life and gear life compared to the use of Cat TDTO or TO-4 oils.

The S·O·S oil analysis program is recommended when extending the oil drain intervals.

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Gear Oil

SMCS Code: 7000; 7551; 7581

NOTICE
Do not use the Cat GO (Gear Oil) or commercial gear oil in the machine compartments unless specifically recommended by Caterpillar. The gear oil can cause seals to fail. The seals can also leak. The gear oil may not be compatible with friction materials. The oil can reduce the efficiency of the transmission and the brake performance.

Applications

Gear oils are recommended for use in certain Direct Drive Transmissions, Differentials, Final Drives, Transfer drives, Circle Drives (excavators), and other applications. The details of the applications of these oils are given in the “Lubricant Viscosity” section of this Special Publication.

Cat GO (Gear Oil)

Cat GO offers maximum protection against the following damage: scoring of the gear teeth, pitting of the gear teeth and pitting of the parts in roller bearings. Cat GO provides excellent stability under high temperature conditions. Cat GO also has superior low temperature performance. This oil also gives protection against rust and corrosion. Some applications require additives for the extreme pressures that can occur at the surfaces/edges of the components. For these applications, Cat GO provides the extra protection.

When the use of gear oil is specified, use Cat GO or Cat Synthetic GO in order to maximize the component life.

NOTICE
Cat GO is not the same as Cat TDTO, and does not meet Cat TO-4 or TO-4M oil Performance Requirements. Cat GO or commercial gear oils should not be used in compartments that specify Cat TO-4 or TO-4M oil.

PROOF

NOTICE
Cat GO is not the same as Cat FDAO, and does not meet the Cat FD-1 oil Performance Requirements. Cat GO or commercial gear oil should not be used in compartments that specify Cat FD-1 oil.

Commercial Gear Oils

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

Gear lubricants are classified by the API service category and by the SAE viscosity grade that is defined in “SAE J306”.

If the Cat GO or Cat Synthetic GO cannot be used, select a commercial oil that meets the “API GL-5”specification.

Note: Non-Caterpillar commercial “API GL-5” gear oils are as a group second choice oils.

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Multipurpose Tractor Oil

SMCS Code: 7000; 7581

Application

Multipurpose Tractor Oils are recommended for use in certain Axles, Final Drives, Implement Steering, Rear Drive Axles, and other applications. The details of the applications of these oils are given in the “Lubricant Viscosity” section of this Special Publication.

Cat MTO is multigrade oil that can be used in cold weather operations. Refer to the “Viscosity Tables” in the “Lubricant Viscosities” section of this Special Publication for details.

Cat MTO (Multipurpose Tractor Oil)

NOTICE
Cat MTO is not the same as Cat TDTO, and does not meet Cat TO-4 or TO-4M transmission/drive train oil specifications. Cat MTO should not be used in compartments that specify Cat TO-4 or TO-4M oil.

NOTICE
Cat MTO is not the same as Cat FDAO, and does not meet the Cat FD-1 oil specification. Cat MTO should not be used in compartments that specify Cat FD-1 oil.

Cat MTO is multigrade oil that is developed, tested and approved by Caterpillar to provide optimal protection for components where approved for use. Cat MTO offers the following service qualties: improved braking and clutching, enhanced control and extended life in clutches and wet brakes, superior anti-wear properties, thermal stability, and oxidation resistance for long service life in severe applications.

Commercial Multipurpose Tractor Oils

Note: Non-Caterpillar commercial multipurpose tractor oils are as a group second choice oils.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

If Cat MTO is not available, use an oil that meets the “Ford/New Holland M2C134-D” specification. The oil must also satisfy the requirements of the following commercial machine systems:

- Multipurpose tractor transmission
- Hydraulic drives on agricultural tractors and on industrial tractors
- Final drives on agricultural tractors and on industrial tractors

Cat Compactor Oil

Cat Compactor Oil is recommended for use in compactor vibrator drums and other applications. The details of the applications of these oils are given in the “Lubricant Viscosity” section of this Special Publication.

Cat compactor oil is a premium PAO (polyalphaolefin) synthetic gear oil with no viscosity improver. This lubricant has an ISO viscosity grade to 220 and a minimum viscosity index of 152. This lubricant can be used in cold weather operations. Refer to the “Viscosity Tables” in the “Lubricant Viscosities” section of this Special Publication for details.

Commercial oil for this application should have a full synthetic base stock with no viscosity improvers and an ISO viscosity grade to 220, and a minimum viscosity index of 150.**

Automatic Transmission Oil

Automatic Transmission Oils are classified by the Cat AT-1 (Automatic Transmission - 1) oil Performance Requirements. Caterpillar developed AT-1 for use in automatic transmissions of on-highway trucks and in Caterpillar CX Series on-highway transmissions.

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Cat ATF-HD2 automatic transmission fluid is the preferred oil to maximize the life and performance of automatic transmissions where recommended for use.

Application

Automatic Transmission oils are recommended for use in automatic transmissions of on-highway trucks and in Caterpillar CX Series on-highway transmissions. Refer to the Operation and Maintenance Manual for information for details.

NOTICE
Do not use Cat TDTO fluids in CX Series on-highway automatic transmissions.

Cat ATF-HD2

Cat ATF-HD2 fluid is synthetic automatic transmission fluid developed to provide optimal performance in Caterpillar CX Series on-highway transmissions. Cat ATF fluid can be used in automatic transmissions that are designed to operate with fluids meeting Cat AT-1 or Dexron-III(H) Performance Requirements. Consult your Caterpillar dealer for Cat ATF-HD2 availability.

Cat ATF-HD2 increases the standard oil drain interval for CX Series on-highway automatic transmission (4 to 6 times) over second choice oils when following the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your particular transmission.

Note: Proper warm up procedures should be followed when the temperature is less than −23° C (−10° F). In order to maintain optimum operating conditions of the bearing components, allow the engine to warm up for about 20 minutes. As an alternative, properly sized transmission heater can be installed. Follow all instructions and consult your Caterpillar dealer for cold weather operation of Cat CX Series on-highway automatic transmissions.

Commercial Fluids

For use in Cat CX Series on-highway automatic transmissions, commercial fluids must comply with the requirements of Cat AT-1 specification or the requirements of Dexron-III(H).

Note: Non-Caterpillar commercial oils that meet the Cat AT-1 Performance Requirements are second choice oils.

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Lubricant Viscosities

SMCS Code: 1000; 7000; 7581

Selecting the Viscosity

Ambient temperature is the temperature of the air in the immediate vicinity of the machine. The temperature may differ due to the machine application from the generic ambient temperature for a geographic region. When selecting the proper oil viscosity for use, review **both** the regional ambient temperature and the potential ambient temperature for a given machine application. Generally, use the higher temperature as the criterion for the selection of the oil viscosity. Generally, use the highest oil viscosity that is allowed for the ambient temperature when you start the machine. Refer to the “Lubricant Viscosities for Ambient Temperatures” tables for guidance. In cold-weather applications, the preferred method is to use properly sized machine compartment heaters and a higher viscosity grade oil. Thermostatically controlled heaters that circulate the oil are preferred.

The proper oil viscosity grade is determined by the minimum ambient temperature (the air in the immediate vicinity of the machine). Ambient temperature is the temperature when the machine is started and while the machine is operated. To determine the proper oil viscosity grade, refer to the “Min” column in the table. This information reflects the coldest ambient temperature condition for starting a cold machine and for operating a cold machine. Refer to the “Max” column in the table to select the oil viscosity grade for operating the machine at the highest temperature that is anticipated. Unless specified otherwise in the “Lubricant Viscosities for Ambient Temperatures” tables, use the highest oil viscosity that is allowed for the ambient temperature when you start the machine.

Machines that are operated continuously should use oils that have the higher oil viscosity in the final drives and in the differentials. The oils that have the higher oil viscosity will maintain the highest possible oil film thickness. Refer to this Special Publication, General Information for Lubricants article, Lubricant Viscosities tables, and any associated footnotes. Consult your Cat dealer if additional information is needed.

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Note: SAE 0W and SAE 5W oils, where allowed for use in non-hydraulic system compartments, are not recommended for use in machines that are operated continuously and/or are heavily loaded. Refer to the “Lubricant Viscosities for Ambient Temperatures” tables for guidance. The oils that have the higher oil viscosity will maintain the highest possible oil film thickness. Consult your Cat dealer if additional information is needed.

Note: Oil viscosity grade selection is also machine compartment specific. Some machine models and/or machine compartments do not allow the use of all available viscosity grades. For guidance on selecting oil viscosity, refer to the “Lubricant Viscosities for Ambient Temperatures” tables.

Note: Generally, use the highest oil viscosity that is available to meet the requirement for the temperature at start-up.

NOTICE
Proper oil viscosity **AND** oil type/specification are required to maximize machine compartment performance and life. Do **NOT** use only oil viscosity, or only oil type to determine the machine compartment oil selection. Using only the oil viscosity or only the oil type to determine a machine compartment oil selection can lead to reduced performance and compartment failure. Refer to the “Lubricant Viscosities for Ambient Temperatures” tables and to ALL of the associated footnotes.

NOTICE
Not following the recommendations found in the “Lubricant Viscosities for Ambient Temperatures” tables and associated footnotes can lead to reduced performance and compartment failure.

NOTICE
In colder ambient conditions a machine warm-up procedure and/or supplemental machine fluid compartment heat may be required. Machine specific warm-up procedures can typically be found in the Operation and Maintenance Manual for the machine. In addition, generic machine warm-up procedures can be found in this Special Publication, “Procedures for Machines that are Used in Cold Weather - (Generic)” topic.

General Information for Lubricants

The information provided in this “Lubricant Viscosities for Ambient Temperatures” article and Tables should be used with the information provided in the “Lubricant Specifications” section (Maintenance Section) of this Special Publication.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids and greases.

NOTICE
Not following the recommendations found in this Special Publication can lead to reduced performance and compartment failure.

NOTICE
Do NOT use only the oil viscosities when determining the recommended oil for an engine compartment. The oil type (performance requirements) MUST also be used.

Note: Some machine models and/or machine compartments do NOT allow the use of all available oil viscosity grades.

Note: Only use the oil type and the specification that is recommended for the various machine compartments.

Note: Some machine compartments allow the use of more than one oil type. For the best results, do not mix oil types.

Note: Different brand oils may use different additive packages to meet the various machine compartment performance specification recommendations. For the best results, do not mix oil brands.

Note: The availability of the various Cat oils will vary by region.

Note: Cat oils are the preferred oils. All other oil types and specifications that are listed in the applicable section are acceptable oils.

The minimum requirements for commercial oils for machine components are provided in Table 6 of this Special Publication. These requirements apply to all the fluids recommendations given in this Special Publication. Commercial oils are, as a group, second choice oils.

The factory fill oils for standard configuration machines are indicated in the lubricant recommendations tables, where pertinent. Factory fill information is general and exceptions may exist. For example, machines that are shipped to cold climates may have other factory fill oils.

When you are operating the machine in temperatures below −20°C (−4°F), refer to Special Publication, SEBU5898, “Cold Weather Recommendations”. This publication is available from your Cat dealer.

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